

Question 1

(a) $L = (\partial_x + x\partial_y)$

$$\begin{aligned}
L(u + v) &= (\partial_x + x\partial_y)(u + v) \\
&= u_x + v_x + xu_y + xv_y \\
&= Lu + Lv \\
L(cu) &= (\partial_x + x\partial_y)(cu) \\
&= cu_x + cxu_y \\
&= c(u_x + xu_y) \\
&= cLu
\end{aligned}$$

Therefore L is a linear operator.

(b) $L = (\partial_x + u_y\partial_y)$

$$\begin{aligned}
L(u + v) &= (\partial_x + u_y\partial_y)(u + v) \\
&= u_x + v_x + (u_y)^2 + u_yv_y \\
&\neq (u_x + (u_y)^2) + (v_x + (v_y)^2) = Lu + Lv
\end{aligned}$$

Therefore, L is NOT a linear operator.

(c) $L = (\partial_x + \partial_y + \frac{1}{u})$

$$\begin{aligned}
L(u + v) &= (\partial_x + \partial_y + \frac{1}{u})(u + v) \\
&= u_x + v_x + u_y + v_y + 1 + \frac{v}{u} \\
&\neq (u_x + u_y + 1) + (v_x + v_y + \frac{v}{u}) = Lu + Lv
\end{aligned}$$

Therefore, L is NOT a linear operator.

(d) $L = (\cos(x)\partial_{xy} + \arctan(\frac{y}{x})\partial_y)$

$$\begin{aligned}
 L(u + v) &= (\cos(x)\partial_{xy} + \arctan(\frac{y}{x})\partial_y)(u + v) \\
 &= \cos(x)u_{xy} + \arctan(\frac{y}{x})u_y + \cos(x)v_{xy} + \arctan(\frac{y}{x})v_y \\
 &= Lu + Lv \\
 L(cu) &= (\cos(x)\partial_{xy} + \arctan(\frac{y}{x})\partial_y)(cu) \\
 &= c\cos(x)u_{xy} + c\arctan(\frac{y}{x})u_y \\
 &= c(\cos(x)u_{xy} + \arctan(\frac{y}{x})u_y) \\
 &= cLu
 \end{aligned}$$

Therefore, L is a linear operator.

Question 2

Question 3

(a)

$$\begin{aligned}
 u_x &= 0 \\
 u &= A(y)
 \end{aligned}$$

(b)

$$\begin{aligned}
 u_x &= x^2 + y \\
 u &= \frac{1}{3}x^3
 \end{aligned}$$

Question 4

(a)

Question 5

(a)

Question 6